

Disproof of Conjecture 00000001128

Maximal Schubert-polynomial coefficients are not a factorial times a lattice-path count

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Conjecture (00000001128), quoted verbatim. “Definition: $M(w, j)$ is the maximal monomial coefficient in the degree- j part of the Schubert polynomial $\mathfrak{S}_w$. Conjecture: $M(w, j)$ equals $\min(j, \ell(w), \lceil \ell(w)/2 \rceil)!$ times a lattice-path count $C(w)$, where $C(w)$ has a product formula on 321-avoiding permutations; in particular $M(w, \ell(w))$ is an explicit function of the increasing-subsequence count of w .”

We disprove the conjecture with a single counterexample. Since $C(w)$ is a *count* of lattice paths, it must be a non-negative integer; the conjecture forces $C(321) = 1/2$.

1 The counterexample $w = 321 \in S_3$

Let $w = 321$ (one-line notation), the longest element w_0 of S_3 . Then $\ell(w) = 3$, and w has longest increasing subsequence length 1.

The Schubert polynomial. For the longest element of S_n one has the product formula

$$\mathfrak{S}_{w_0} = \prod_{1 \leq i < j \leq n} (x_i + x_{i+1} + \cdots + x_{j-1}),$$

so in $n = 3$ variables

$$\mathfrak{S}_{321} = x_1(x_1 + x_2)x_2 = x_1^2x_2 + x_1x_2^2.$$

This polynomial is homogeneous of degree 3 (equivalently, its degree-3 part is the whole polynomial), so there is no ambiguity in the notion of “degree- j part”.

The maximal coefficient. Both monomials $x_1^2x_2$ and $x_1x_2^2$ of the degree-3 part carry coefficient 1; hence

$$M(321, 3) = 1. \tag{1}$$

This value was confirmed by an independent exact computation of $\mathfrak{S}_w$ via the down-transition recurrence

$$\mathfrak{S}_{\text{id}} = 1, \quad \mathfrak{S}_w = \sum_{i: \ell(wt_i) < \ell(w)} x_i \cdot \mathfrak{S}_{wt_i},$$

implemented in `reproduce.py` in exact integer arithmetic; the resulting polynomial $x_1^2x_2 + x_1x_2^2$ agrees with the product formula above.

The conjectured factorial factor. With $j = 3$ and $\ell(w) = 3$,

$$\min(j, \ell(w), \lceil \ell(w)/2 \rceil)! = \min(3, 3, \lceil 3/2 \rceil)! = \min(3, 3, 2)! = 2! = 2. \quad (2)$$

2 The contradiction

Suppose the conjecture holds at $(w, j) = (321, 3)$. Then by (1) and (2)

$$1 = M(321, 3) = 2 \cdot C(321),$$

with $C(321)$ a lattice-path count, i.e. a non-negative integer. But $2 \nmid 1$, so the unique rational solution is $C(321) = \frac{1}{2} \notin \mathbb{Z}_{\geq 0}$, which cannot be the cardinality of any set of lattice paths. Contradiction.

The failure is generic in nature: whenever $\ell(w)$ is odd and the conjectured factor $\min(j, \ell(w), \lceil \ell(w)/2 \rceil)!$ contains an even factor at $j = \ell(w)$, every odd value of $M(w, j)$ is immediately excluded — and odd maximal coefficients occur already at the top element of S_3 . One counterexample suffices.

Claim. *Conjecture 00000001128 is false: the identity $M(w, j) = \min(j, \ell(w), \lceil \ell(w)/2 \rceil)! \cdot C(w)$ fails at $w = 321$, $j = 3$, since it would force $C(321) = 1/2$, which is not a lattice-path count.*

Artifacts.

- `reproduce.py` — exact recomputation of $\mathfrak{S}_{321}$ (down-transition recurrence, cross-checked against the product formula), extraction of $M(321, 3)$, the factor of (2), and the divisibility failure.
- `lean4/Main.lean` — a zero-axiom Lean 4 certificate (no `sorry`, no `propext`, no `Classical.choice`, no `Quot.sound`) proving $\ell(321) = 3$, $M(321, 3) = 1$, and $\neg \exists C : \mathbb{N}, M(321, 3) = k! \cdot C$ for $k = \min(3, 3, \lceil 3/2 \rceil)$.